

Design, Fabrication, Testing, and Delivery of Custom Carbon Fiber Propellers

STATEMENT OF WORK

1. INTRODUCTION

This Statement of Work (SoW) describes the requirements for the design, fabrication, structural testing, and delivery of a custom carbon fiber propeller. The propeller will be tested at the NASA Icing Research Tunnel (IRT) to evaluate icing scaling methods for propellers as part of an international Space Act Agreement (SAA).

1.1. SCOPE

The contractor shall provide two (x2) carbon fiber propellers, (x1) variable pitch hub, (x1) spinner, (x1) overspin test report, and (x1) static centrifugal load test report.

2. REQUIREMENTS

- 2.1.1. The contractor shall provide two (x2) identical carbon fiber propellers of the following specifications:
 - 2.1.1.1. The propeller shall have 3 blades each which are individually removable from the hub.
 - 2.1.1.2. The propellers shall be 1 meter in diameter.
 - 2.1.1.3. The propellers shall have custom twist and chord distributions which will be provided by NASA.
 - 2.1.1.4. The “as-manufactured” propeller geometry should match the “as-designed” propeller geometry by ± 0.004 ” (± 0.1 mm).
 - 2.1.1.5. The propellers shall be designed to safely operate at a target max RPM of 3500 RPM while achieving a safety factor of at least 1.5.
 - 2.1.1.6. The propeller shall be made of carbon fiber.
 - 2.1.1.7. Each individual propeller blade shall not exceed 1 lb.
- 2.1.2. The contractor shall provide one (x1) CAD model of the final propeller design. This will be considered the “as-designed” propeller geometry.
- 2.1.3. The contractor shall provide one (x1) variable pitch hub of the following specifications:
 - 2.1.3.1. The hub shall be a 3-bladed hub.
 - 2.1.3.2. The hub shall allow the user to adjust the blade pitch angle in between tests (ground-adjustable pitch).
 - 2.1.3.3. The hub shall allow the user to “lock” the blade pitch angle at five discrete pitch angles. The five pitch angles will be specified by NASA.

- 2.1.3.4. The blade pitch angle shall not deviate more than $\pm 0.1^\circ$ from the target pitch setting for the five discrete pitch angles mentioned in 2.1.3.3.
- 2.1.3.5. The hub shall allow attachment to an EMRAX 188 electric motor. The mounting pattern for the EMRAX 188 motor will be provided by NASA.
- 2.1.3.6. The hub shall be designed to safely operate at a target max RPM of 3500 RPM (with the propellers installed) while achieving a safety factor of at least 1.5.
- 2.1.4. The contractor shall provide one (x1) CAD model of the variable pitch hub.
- 2.1.5. The contractor shall provide one (x1) spinner of the following specifications:
 - 2.1.5.1. The spinner shall attach to the variable pitch hub.
 - 2.1.5.2. The spinner shall have a diameter of 188 mm.
 - 2.1.5.3. The spinner geometry will be provided by NASA.
 - 2.1.5.4. The spinner shall be made of carbon fiber.
- 2.1.6. The contractor shall provide one (x1) CAD model of the spinner.
- 2.1.7. The contractor shall provide a test report for an overspin test conducted on the propeller to verify that it can safely operate at the target max RPM.
 - 2.1.7.1. The propeller blades shall be installed in the variable pitch hub during the overspin test.
 - 2.1.7.2. The overspin test shall be performed at a speed 20% higher than the target max RPM for a duration of 30 mins.
- 2.1.8. The contractor shall provide a test report for a centrifugal load tensile test conducted on a propeller blade.
 - 2.1.8.1. The propeller blade to be tested shall be installed in the variable pitch hub to verify that the propeller fails prior to the hub.
 - 2.1.8.2. The tensile test shall stress the propeller blade in the spanwise direction.
 - 2.1.8.3. The tensile test shall be conducted to failure to determine the max centrifugal load the propeller (or hub) can withstand.

3. SHIPPING RESPONSIBILITIES

The government shall pay for all shipping costs associated with shipping the item(s) from the Contractor facility.

4. DELIVERABLES

1. (x2) Carbon fiber propellers
2. (x1) Variable pitch hub
3. (x1) Spinner
4. (x1) Overspin test report
5. (x1) Static centrifuge load test report

5. DELIVERY SCHEDULE

All deliverables shall be completed within 3-4 months from the date of contract award.

6. NASA POINT OF CONTACT

Please direct questions and correspondence to:

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